POWER & COMPATIBILITY SHEET

SMARTSTOCK SYSTEM ARCHITECTURE • HARDWARE & SENSORS

STATUS: **VERIFIED**
LOGIC: **3.3V DC**

GPIO CEILING	MAIN POWER RAIL	SENSOR BUS	RELAY BUS
3.3V	**5V / 3A**	**Phidget VINT**	**Opto-5V**
Strict Limit (RPi / Arduino)	Dedicated USB-C Supply	Isolated Hub Management	Optocoupler Isolated

1. ELECTRICAL COMPATIBILITY MATRIX

TBL-01

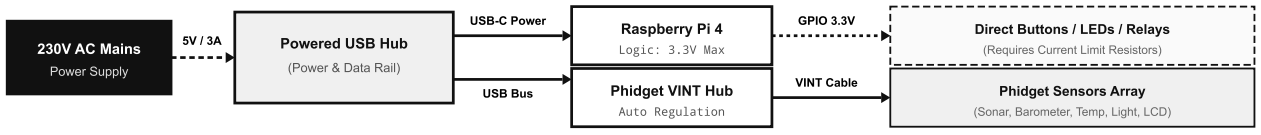
COMPONENT	POWER RAIL	LOGIC LEVEL	CONSTRAINT & INTERFACING RULE
Raspberry Pi 4	5V DC (USB-C 3A)	3.3V MAX	Inputs are non-5V tolerant. Direct 5V connections will damage the SoC.
Arduino 1010	5V DC (MicroUSB/Vin)	3.3V MAX	Microcontroller I/O operates at 3.3V. Requires logic shifter for 5V signals.
Phidget VINT Hub	5V DC (USB Bus)	Hub Managed	ISOLATED Auto-regulates voltages for all connected Phidget sensors.
Phidget Sensors	Via VINT Hub	3.3V / 5V Auto	PLUG & PLAY Integrated signal conditioning via proprietary cable.
Dual Relay Board	5V DC (Vcc Coil)	3.3V / 5V Opto	COMPATIBLE Control signal accepts 3.3V GPIO through onboard optocouplers.
LEDs / Switches	3.3V or 5V DC	Discrete I/O	Series resistor mandatory (I_{max} ≤ 16mA per GPIO channel).

2. SYSTEM POWER DISTRIBUTION TOPOLOGY

FIG-01

BLOCK DIAGRAM — POWER ISOLATION & SIGNAL PATHS

SCALE: NTS



3. CRITICAL OPERATING & SAFETY DIRECTIVES

SEC-01

[RULE 1] GPIO VOLTAGE THRESHOLDS

The GPIO pins on both the **Raspberry Pi 4** and **Arduino 1010** operate at a strict **3.3V logic ceiling**. Interfacing 5V sensors directly will result in permanent chip failure. A bidirection logic level converter is required for all non-isolated 5V digital signals.

[RULE 2] POWER ISOLATION & BROWNOUT PREVENTION

To prevent system brownouts caused by transient current spikes from USB peripherals, power is routed through an **External Powered USB Hub**. This insulates the Raspberry Pi 4 power plane while providing steady current to the Phidget VINT Hub and interface boards.